ID5059: Summary Report

The nycflights13 dataset comprises flight information from 3 major airports (EWR, JFK, LGA) near New York over the course of 2013. Across the aviation industry, an airline arrival occurring 15+ minutes after the scheduled time is considered delayed, so the same assumption has been used here. We also assume that historical flights delay patterns are indicative of future flight delays. Delays were investigated using offline supervised binary classification machine learning models.

Can we predict before a flight has departed, whether it will arrive late?

To predict whether a flight would arrive late I built 4 initial models:

- Logistic Regression Classifier (LR)
- Stochastic Gradient Descent Classifier (SGD)
- Random Forest Classifier (RF)*
- Histogram-based Gradient Boosting Classifier (HGB)

* Random forest initially overfit the training data so it was tuned slightly for comparison but not used as the final model.

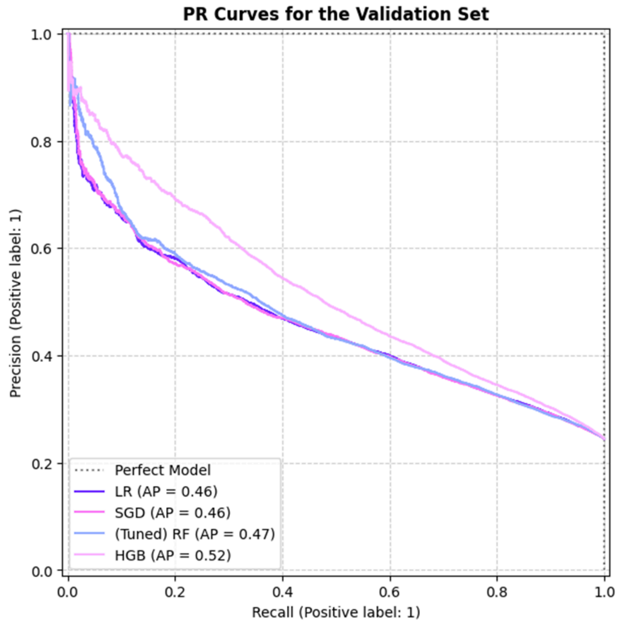
When evaluating the models, <u>recall</u> was prioritised over <u>precision</u>. Recall measures how many actual delays a model correctly identifies, while precision measures how many predicted delays were correct. I argue that missing a delay (false negative) is more costly than over-predicting delays (false positives), because if our airport doesn't account for enough delays it may heavily disrupt operations and reduce passenger satisfaction versus the smaller cost of over-preparing for delays.

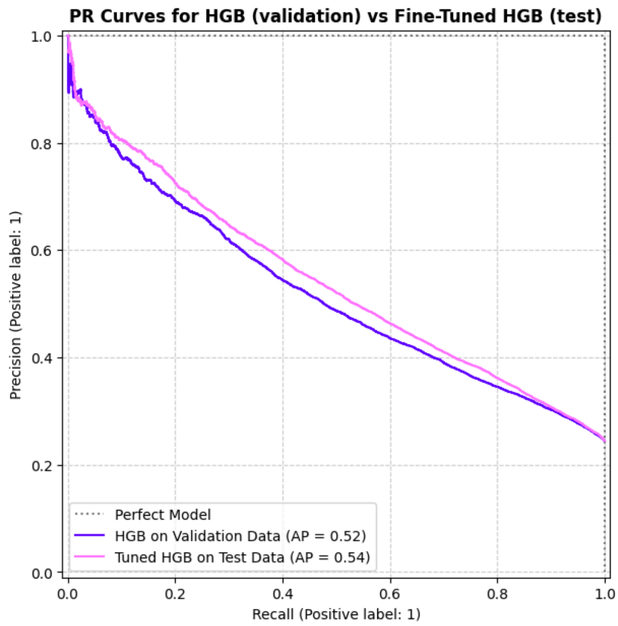
PR curves were primarily used for evaluating model performance as they are more reliable for imbalanced datasets like nycflights13 where ~25% of flights were delayed and ~75% were on-time.

Fine-Tuned Model: HGBClassifier

SGD had the best recall overall, identifying 68% of all real delays (vs HGB's 65%). However, HGB performed best overall on unseen validation data with an F1-score of 0.51, showing a more robust precision/recall balance than SGD (0.48). This is shown by HGB's line being closer to the "perfect model" on the validation set line chart, and why it was picked for fine tuning.

After tuning hyperparameters such as how greedily the model learns (learning rate), how complex rules can be (max tree depth), and how many flights are needed to make a rule (min samples leaf), HGB achieved a recall of 0.66 and F1-Score of 0.52 on unseen test data. Therefore, **we can correctly predict around two thirds (66%) of delayed flights** while maintaining reasonable precision.





Final Model Recall: 0.6582 **Final Model F1:** 0.5224

Class	Precision	Recall	F1-Score	Support
"On Time"	0.87	0.72	0.79	49447
"Delayed"	0.43	0.66	0.52	16023
"Accuracy"	—	—	0.71	65470
"MacroAvg"	0.65	0.69	0.65	65470
"Weighted Avg"	0.76	0.71	0.72	65470

Which factors are most correlated with delays?

A few key factors were strongly correlated with delays.

Precipitation type had the clearest relationship. Snow conditions caused a ~77% chance of delayed flights, followed by rain and then clear weather. These weather conditions were derived from the temperature (above/below freezing) and the presence of precipitation.

Visibility also showed a strong relationship. Low visibility corresponded to a ~45% delay rate, decreasing across medium (haze) and high (clear) visibility conditions.

Beyond weather, the carrier (airline) also influenced delay probability. The purple bar chart shows substantial variation across airlines: F9 ("Frontier Airlines Inc") had the highest delay rate of ~40%, while HA ("Hawaiian Airlines Inc") had the lowest at ~10%.

Chronological features were also important. Delay rates increased as the hour of the day progressed, likely reflecting compounding knock-on delays since the number of flights operating throughout the day was fairly consistent (so delays later can't be caused purely by an increase in flights in the evening, for example). The month also mattered, with June, July, and December showing peak delay rates. June/July may reflect Atlantic hurricane season, while December may relate to increased holiday traffic, with nearly double the number of flights in December compared to all other months.

Remark: Feature Importances

A machine-learning model is a black-box, so we can inspect "Feature Importances" to see which factors it places the most importance on.

While the aforementioned factors were most correlated with delays, they weren't always the factors most used by the model for making predictions. Dew point had the highest importance (0.07), followed by weather conditions, wind speed, and then visibility, month, and hour. This is likely due to the fact that although precipitation and visibility had the strongest relationships with delays, the percentage of data with low visibility, rain, or snow was very small overall.

That is to say there aren't many snowy or foggy days in New York, but when there are, a flight is very likely to be delayed.